Ashish Tummurugoti

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Personal Statement

MSc Motorsport Engineering graduate from Oxford Brookes University with three years of Formula Student and one year of industry experience in vehicle dynamics, modelling and simulation. Experienced in developing and validating vehicle models using MSC Adams Car, AVL VSM and IPG CarMaker, with programming skills in C++, Python and MATLAB. Seeking graduate opportunities in vehicle dynamics, modelling and driver-in-the-loop simulation.

Work Experience

Vehicle Dynamics Member, Oxford Brookes Racing – Oxford, United Kingdom Sep 2025 – Sep 2026

- Validated suspension kinematics and transient response to within 10% between a full-vehicle MSC Adams Car model and AVL VSM, by correlating lateral acceleration, load transfer and compliance results.
- Cut virtual lap time by 3.5 s against the baseline IPG CarMaker model, by running sensitivity sweeps of powertrain, aerodynamic and suspension parameters, refining the driver model and documenting the findings in a technical report.
- Enabled repeatable suspension stiffness measurements to validate simulation models, by designing and fabricating test jigs with the vehicle dynamics and manufacturing teams.

Project Intern, Coexlion – Bengaluru, India Apr 2025 – Aug 2025

- Increased EV two-wheeler wheel travel by 20 mm to address client concerns about ride comfort over bumps, optimising suspension motion ratio and spring-damper placement within fixed chassis and damper eye-to-eye constraints.
- Reduced gear-shift time by 2.08 s, by building, testing and iterating an Arduino-based automated gear-shifting and clutch-actuation prototype that integrated mechanical and electronic systems.
- Supported suspension design decisions, by building a Python-driven Excel kinematics and force solver that reads geometry inputs, calculates results across configurations, verifies them against hand calculations and generates graphs.

Vehicle Dynamics Member, Vega Racing Electric – Bengaluru, India Feb 2023 – Feb 2025

- Improved low-speed corner rotation to an understeer gradient of -1.5°/g, by iteratively tuning dampers at the track and calculating the gradient from the trend of logged steering angle and lateral acceleration data.
- Increased usable suspension wheel travel by 15% over the baseline to increase bump absorption and tyre contact, by optimising the kinematics to a 0.73 motion ratio and 0.03°/mm camber gain.

Project Intern, Gas Turbine Research Establishment (DRDO) – Bengaluru, India Mar 2024 – Jul 2024

- Identified the critical structural regions that set the design requirements for a new composite engine casing, by simulating 12 pressure, actuator, jet-reaction and combined load cases in Abaqus.
- Improved casing stiffness-to-weight ratio by 12% over the baseline while holding a 2.5 safety factor under 3g and 4.5g loading, by optimising composite ply orientation and material properties.

Project Intern, Aeronautical Development Establishment (DRDO) – Bengaluru, India Jun 2023 – Aug 2023

- Predicted a burnout velocity of 71.94 m/s for an 80 kg aircraft and 20 kg booster, by modelling 6-DOF multi-body separation dynamics in MSC Adams with joint and contact modelling.
- Achieved an 8.3 mm clearance from the engine bay and 43.45 mm from the fuselage pin holder, by running 1,000+ simulations with custom spring-force modelling.
- Validated the MSC Adams model against hand calculations to within 5%, by refining force functions, contact parameters and model inputs through iterative simulation.

Data Engineer Intern, Prudent Motorsport – Bengaluru, India May 2022 – Jul 2022

- Improved average lap time by 20 s across three drivers over a week of testing, by producing driver performance sheets from telemetry and track data for three drivers and reviewing them in debrief meetings.
- Analysed lateral acceleration, throttle and brake telemetry in AIM Race Studio to evaluate driver/vehicle performance.
- Supported the setup crew at 12 track sessions, by checking tyre pressures and temperatures and making setup changes.

Projects

MSc Dissertation: MBD Analysis of Conventional vs. Mode-Decoupled Suspension Mar 2026 – Sep 2026

- Quantified tyre load variation, lateral load transfer distribution, roll-centre migration and vehicle behaviour, by developing a full-vehicle MSC Adams Car model with a PAC2002 tyre model and a C++ yaw moment diagram tool.
- Achieved kinematic and suspension response error below 12%, by cross validating the Adams Car model against the AVL VSM kinematic solver.
- Reduced roll angle by 20% and lateral load transfer distribution by 6%, by applying a passive mode-decoupled suspension architecture that controls heave and roll independently.

2026 F1 Vehicle Development and Race Strategy Simulation Jan 2026 – May 2026

- Collaborated in a team of six to establish the 2026 F1 development baseline by adapting a 2025 car model to the new regulations, updating vehicle mass, wheelbase, track width, active aerodynamics and powertrain parameters.
- Cut lap time by 0.95 s, by analysing AVL VSM simulation data in DRIVE Race and MATLAB with yaw moment and G-G diagrams to identify aero-induced understeer and resolving it through rear suspension optimisation.
- Reduced lap time by 1.02 s, by iterating vehicle dynamics simulations of weight distribution, spring/ARB stiffness and camber/toe to identify an optimal 44:56 front-to-rear balance.
- Improved lap time by 0.67 s, by quantifying tyre degradation and fuel-mass sensitivity from tyre, fuel and energy-management data and optimising MGU-K deployment sequencing for race strategy.

Vehicle Dynamics Modelling and Ride Optimisation for a Formula Student Car Sep 2025 – Dec 2025

- Validated bicycle and twin-track MSC Adams Car models to within 1.4% of published skid-pad results, by using nonlinear tyre modelling and lateral-acceleration correlation.
- Cut peak pitch angle by 19%, by running a 250+ trial design of experiments on spring and damping rates in 2DOF and 4DOF Adams models with four-post rig inputs and nonlinear damper data.
- Lowered peak front outer-tyre load transfer by 12%, by comparing 100% Ackermann and anti-Ackermann steering geometries in a twin-track step-steer simulation.

Skills

Programming and Data Analysis: C++, Python, MATLAB, AIM Race Studio, Simulink, Arduino
Vehicle Dynamics Simulation: MSC Adams, AVL VSM, IPG CarMaker, DRIVE Race
FEA and Analysis: HyperMesh, Abaqus
Mechanical Design and CAD: SolidWorks

Education

Master of Science in Motorsport Engineering — Oxford Brookes University, Oxford, UK Sep 2026
(Distinction Expected)
Relevant Modules: Vehicle Dynamics, Lap Time Simulation and Race Engineering

Bachelor of Technology in Mechanical Engineering — PES University, Bengaluru, India Dec 2024
(2:1 Equivalent)
Relevant Modules: Engineering Mathematics, Autonomous Vehicles, Control Engineering